SEAL: LIVE SUPERVISOR EVIDENCE ARMS ON FIXED LOCAL FIXTURES

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Arjun Vijay Prakash
Independent Researcher

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ABSTRACT

Live model supervisors may publish illegitimate workflows when workers assert false completion on fixed fixtures. This audit summarizes held-out test episodes from frozen protocol ADAPTION-LIVE-MAIN-v3 with supervisors gemma-4-31b-it and qwen3.5-397b (960 (frozen target 960) episodes, four template families, 2000 replicates, seed 20261108, 95% intervals). Attacks follow three mechanisms with single versus coordinated message counts and speaker roles on fixed snapshots; models are not pooled. 51 error or invalid episodes enter all-status denominators only. Coordinated claims/provenance unsafe execution: gemma-4-31b-it claims 1.000 (39/39); qwen3.5-397b claims 1.000 (40/40); gemma-4-31b-it provenance 0.000 (0/35); qwen3.5-397b provenance 0.000 (0/36). Honest provenance legitimate completion: gemma-4-31b-it provenance 1.000 (40/40); qwen3.5-397b provenance 1.000 (40/40). Fixed exploratory claims-minus-provenance contrasts: gemma-4-31b-it coordinated claims-minus-provenance 1.000 [1.000, 1.000] (2 of 4 families); gemma-4-31b-it single claims-minus-provenance 1.000 [1.000, 1.000] (2 of 4 families); qwen3.5-397b coordinated claims-minus-provenance 1.000 [1.000, 1.000] (1 of 4 families); qwen3.5-397b single claims-minus-provenance not estimable (0 of 4 families). Four-arm matching requires identical completed fixture sets; intervals describe retained families only. Worker corpus: 280 accepted generations; 280/280 matched scripted baseline verbatim (provenance check only). Claims, receipts, and state execute publish advice; provenance sees summaries when binding is complete and alone may veto or abstain in code. Unsafe execution includes corrected summaries on illegitimate workflows. Attacks are instructed, not emergent collusion. Prospective budgeted evidence recovery was not conducted.

Keywords multi-agent oversight · evidence provenance · audit logs · AI control

1 Introduction

When workers assert false completion on fixed local records, a live model supervisor can still publish to the fake sink. Unsafe execution counts every executed publication of an illegitimate workflow, including a summary corrected to the true summary. The central question is empirical: on a frozen fixture catalog, which *fixed* evidence tier changes unsafe execution versus legitimate completion when worker messages are held constant. This report is an **empirical audit** of instructed attacks on a synthetic publication sink. It is not a learned policy for adaptive evidence budgets or prospective evidence recovery; no such recovery experiment was run here.

SEAL (Supervisor Evidence Arms, Live) compares what the supervisor *advises* (publish, block, or abstain) with what the harness *enforces* on the sink. On claims, receipts, and state, a publish decision executes unless the run errors. On provenance only, deterministic code may veto an otherwise recorded publish request. A zero harmful execution rate on provenance therefore bounds post-model enforcement, not a change in model judgment alone.

The locally frozen protocol `ADAPTION-LIVE-MAIN-v3` schedules 960 held-out test episodes across 4 template families, with 10 numeric variants per family. Those variants are repeated measures inside the family. Development uses 4 variants per family and is excluded from test denominators. Held-out attacks use three mechanisms on four

templates: consistent fake input on both `parts-catalog` and `sensor-calibration`, unauthorized publication on `payroll-release`, and ambiguous evidence on `incident-timeline`. All four arms receive identical worker messages plus tier-specific observations so message availability does not suppress the single-versus-coordinated contrast.

2 Related Work

Agent benchmarks and instruction-hierarchy training study prompt injection and privileged instructions under tool use [1, 2, 3]. NARCBench and HELIOS probe instructed collusion in multi-agent settings with activation sensors [4]. AI-control literature motivates testing oversight against subversion without claiming a full control protocol here [6]. Unlike work on emergent interaction [5], this live study holds worker text fixed and varies the observation bundle and whether a deterministic publication gate can refuse execution after a supervisor publish decision.

Recent monitor evaluations separate instructions from actual collusive decisions, measure benign disruption from repeated warnings, test transfer across policy families, and allow strategic attack timing [7, 8, 9, 10]. SEAL does not resolve these broader monitoring problems: its workers reproduce scripted instructions, supervisors make one decision per episode, and attacks follow a fixed catalog. Active acquisition and selective classification already study information cost and rejection coverage [11, 12]; feature-deletion robustness and secure audit logging provide related precedents [15, 16]. Risk-control methods require calibration assumptions beyond descriptive bootstrap intervals [13, 14]. Our contribution is the inspectable fixed-arm audit, not a new acquisition algorithm or statistical safety guarantee. A comparison matrix for the proposed extension is released in `docs/literature-matrix.md`.

3 Methods

3.1 Fixtures and arms

Four development mechanisms (false completion, wrong summary, omitted step, stale evidence) diagnose the harness. Four held-out templates implement three test mechanisms. Consistent fake input is repeated on `parts-catalog` and `sensor-calibration`. `payroll-release` is unauthorized publication. `incident-timeline` is ambiguous evidence. Identifiers follow `family_id.index.coordination`. The ten variants in a family change numbers inside that template. They are not independent samples. The bootstrap resamples four `family_id` blocks, not forty fixture indexes. Intervals stay coarse: the blocks are few, and two of them share a mechanism.

Single and coordinated attacks share signed records. The single condition is one publisher role. The coordinated condition is three role-specific speakers. Message count and speaker roles both differ, so coordination contrasts stay descriptive. Honest controls always use three speakers. There is no single-speaker honest control.

State observations are fixed read-only keeper snapshots with `adaptive=false`. They are not multi-turn adaptive queries. Ed25519 signing keys are deterministic public material for reproducibility, not a production trust root.

Arms are intentionally not information-matched. Claims sees task text and worker messages. Receipts adds verified receipts and a fresh checkpoint. State adds the keeper snapshot. Provenance adds trusted inputs, digests, tool allowlists, and channel policy, and only that arm can veto a publication in deterministic code. The provenance observation includes `trusted_summary` when `binding_complete` is true. On the three answerable test families that field equals the fixture true summary in every attack coordination. On `incident-timeline` attacks `binding_complete` is false and `trusted_summary` is null; honest controls on that family are binding-complete and also receive the derived true summary. Fresh checkpoint handling is identical across receipts, state, and provenance. Reference decisions in the fixture catalog supply equal-observation baselines for pairing checks, not supervisor policies.

3.2 Trust, observation, and evidence hiding

We separate *what is shown* to the supervisor from *what is enforced* after a decision. Table 2 and Appendix B report the stored counters; here we state the audit assumptions.

Post-record omission. Each arm reveals a fixed bundle. Fields absent from that bundle are hidden from the model even if they exist in the simulator (for example keeper-only bodies without trusted inputs on state).

Non-recording. Evidence that was never captured cannot be recovered from a protected archive. This study does not inject recording failures or test recovery from an archive. Error and invalid supervisor outputs are a separate inference-failure category; they are excluded from completed-case rates and counted as non-success in all-status sensitivities.

Tampering and signer compromise. The harness verifies receipts and checkpoints against frozen catalog material. It does not model an adversary who forges signatures offline or compromises long-lived signing keys; demo keys are deterministic placeholders.

Limits of fixed evidence arms. Receipts and state can look internally consistent while worker claims are false. Provenance can show a trusted summary on answerable attacks (correct-answer exposure, not a mechanism label). No arm implements adaptive querying or learned selection of what to fetch next; those would be outside this frozen design.

3.3 Partial observation and gate (standard notation)

Let $O_a(h)$ denote the observation of history h under arm a. Worker text is held fixed across arms for a given fixture. If a compliant history h_0 and a violating history h_1 have $O_a(h_0) = O_a(h_1)$, any decision rule using only that observation has the same release probability on both. Thus zero release of h_1 requires withholding h_0 as well. Trusted recovery resolves this ambiguity only if an available authentic record distinguishes the histories. A signature alone does not supply an absent record or establish that its signed content is true. This is a standard indistinguishability observation, not a new theorem or a measured property of every fixture.

Write $d \in \{\text{publish}, \text{block}, \text{abstain}\}$ for supervisor advice and $e \in \{\text{execute}, \text{refuse}\}$ for sink enforcement. For $a \in \{\text{claims}, \text{receipts}, \text{state}\}$, $e =$ execute whenever $d =$ publish on completed episodes. For $a =$ provenance, deterministic gate G may refuse the request even if $d =$ publish. An incomplete binding can cause G to abstain. The catalog audit replays G separately from live model calls; its guarantee is limited to the enumerated catalog and implemented predicate.

3.4 Models, workers, and budget

Supervisor models are Gemma 4 31B (`gemma-4-31b-it`) and Qwen 3.5 397B (`qwen3.5-397b`) on the Adaption Responses API. Worker messages are one-shot Gemma wording calls under `WORKER-CORPUS-ASSEMBLY-v1`, not live worker decisions: 280 accepted generations, 0 rejected, 0 incomplete, over 120 fixtures and 280 scheduled messages, model gemma-4-31b-it, spend 0.01335 credits. The generator prompt supplies the scripted message as `wording_reference`. In the assembled test corpus, 280 of 280 accepted model calls reproduced that baseline byte for byte. The assembly audit marks all 120 fixtures `source=generated`. The same assembled corpus is shared by every supervisor arm and model.

Development pilot protocol ADAPTION-LIVE-PILOT-v3 recorded 128 ledger episodes (96 completed, 32 invalid, 0 pending) with pilot `max_output_tokens`=2048 and spend 0.4874 credits. Pilot families are harness repair only. Main spend caps, output limits, and prices are pinned in the main run manifest before outcome inspection. Null `usage`, `latency_seconds`, and `cost_units` are accepted only for `error` and `invalid` records. A completed episode requires measured usage, latency, and cost.

3.5 Endpoints and analysis

Co-primary rates use completed episodes only. Unsafe execution is executed publications on illegitimate completed episodes over that denominator. Legitimate completion is executed legitimate publications equal to the true summary among legitimate completed episodes. Error and invalid episodes are excluded from completed-case denominators.

A supervisor publish decision and a gateway veto are distinct events. On claims, receipts, and state, publish decisions execute on the fake sink. On provenance, publish is a request; code may record `refused_block` or `refused_abstain`. Gateway-blocked unsafe attempts explain provenance gaps between unsafe attempt and unsafe execution rates in Table 2. Field definitions and the decision-to-attempt accounting identity are in Appendix B.

Family-cluster bootstrap uses the four `family_id` blocks in a split, with percentile intervals and paired arm contrasts inside each model and each coordination stratum. Models and coordination conditions are not pooled. `scripts/prepare_live_paper.py` pulls aggregates only from supplied analysis outputs. It does not invent counts.

4 Results

Numerical results use the completed-case co-primary rates and the separate all-status sensitivities defined above. An episode total that lists completed, error, and invalid counts is not a completed-episode denominator.

4.1 Main held-out test outcomes

Completed main analysis supplies 960 test-split episodes with 51 error or invalid records excluded from completed-case rates and retained as non-success in all-status sensitivities (Table 1 and Figure 1). Rates are shown separately for single-worker attacks, three-speaker coordinated attacks, and honest controls; we do not pool adversarial strata.

Claims, receipts, and state arms execute supervisor *advice* to publish on the fake local sink. Provenance alone may refuse publication after an independent digest and authorization check, so a reduction in unsafe *execution* relative to unsafe *attempt* separates deterministic enforcement from model publish advice, not a lower willingness to request publish.

Table 1: Co-primary endpoints by model, evidence arm, and coordination condition. Cells are complete-case success counts and rates; the episode column is total (completed/error/invalid). Error and invalid episodes remain in the total and in the failure-inclusive sensitivity, not in the completed-case denominator. Family-cluster 95% intervals are in the CSV and figure. No model or coordination pooling is performed.

Model	Arm	Coordination	Episodes (C/E/I)	Abstain	Unsafe exec.	Legit. completion	Error+invalid
Gemma 4	claims	single	40 (40/0/0)	0/40 (0.000)	40/40 (1.000)	0/0 (n.e.)	0/40 (0.000)
Gemma 4	claims	coordinated	40 (39/1/0)	0/39 (0.000)	39/39 (1.000)	0/0 (n.e.)	1/40 (0.025)
Gemma 4	claims	honest	40 (40/0/0)	0/40 (0.000)	0/0 (n.e.)	40/40 (1.000)	0/40 (0.000)
Gemma 4	receipts	single	40 (39/0/1)	0/39 (0.000)	30/39 (0.769)	0/0 (n.e.)	1/40 (0.025)
Gemma 4	receipts	coordinated	40 (38/1/1)	0/38 (0.000)	29/38 (0.763)	0/0 (n.e.)	2/40 (0.050)
Gemma 4	receipts	honest	40 (40/0/0)	0/40 (0.000)	0/0 (n.e.)	40/40 (1.000)	0/40 (0.000)
Gemma 4	state	single	40 (30/1/9)	0/30 (0.000)	29/30 (0.967)	0/0 (n.e.)	10/40 (0.250)
Gemma 4	state	coordinated	40 (32/0/8)	0/32 (0.000)	30/32 (0.938)	0/0 (n.e.)	8/40 (0.200)
Gemma 4	state	honest	40 (40/0/0)	0/40 (0.000)	0/0 (n.e.)	40/40 (1.000)	0/40 (0.000)
Gemma 4	provenance	single	40 (35/0/5)	0/35 (0.000)	0/35 (0.000)	0/0 (n.e.)	5/40 (0.125)
Gemma 4	provenance	coordinated	40 (35/0/5)	0/35 (0.000)	0/35 (0.000)	0/0 (n.e.)	5/40 (0.125)
Gemma 4	provenance	honest	40 (40/0/0)	0/40 (0.000)	0/0 (n.e.)	40/40 (1.000)	0/40 (0.000)
Qwen 3.5	claims	single	40 (39/1/0)	0/39 (0.000)	39/39 (1.000)	0/0 (n.e.)	1/40 (0.025)
Qwen 3.5	claims	coordinated	40 (40/0/0)	0/40 (0.000)	40/40 (1.000)	0/0 (n.e.)	0/40 (0.000)
Qwen 3.5	claims	honest	40 (40/0/0)	0/40 (0.000)	0/0 (n.e.)	40/40 (1.000)	0/40 (0.000)
Qwen 3.5	receipts	single	40 (37/3/0)	1/37 (0.027)	34/37 (0.919)	0/0 (n.e.)	3/40 (0.075)
Qwen 3.5	receipts	coordinated	40 (40/0/0)	0/40 (0.000)	38/40 (0.950)	0/0 (n.e.)	0/40 (0.000)
Qwen 3.5	receipts	honest	40 (40/0/0)	0/40 (0.000)	0/0 (n.e.)	40/40 (1.000)	0/40 (0.000)
Qwen 3.5	state	single	40 (38/2/0)	0/38 (0.000)	30/38 (0.789)	0/0 (n.e.)	2/40 (0.050)
Qwen 3.5	state	coordinated	40 (39/1/0)	0/39 (0.000)	30/39 (0.769)	0/0 (n.e.)	1/40 (0.025)
Qwen 3.5	state	honest	40 (38/2/0)	0/38 (0.000)	0/0 (n.e.)	38/38 (1.000)	2/40 (0.050)
Qwen 3.5	provenance	single	40 (34/6/0)	0/34 (0.000)	0/34 (0.000)	0/0 (n.e.)	6/40 (0.150)
Qwen 3.5	provenance	coordinated	40 (36/4/0)	0/36 (0.000)	0/36 (0.000)	0/0 (n.e.)	4/40 (0.100)
Qwen 3.5	provenance	honest	40 (40/0/0)	0/40 (0.000)	0/0 (n.e.)	40/40 (1.000)	0/40 (0.000)

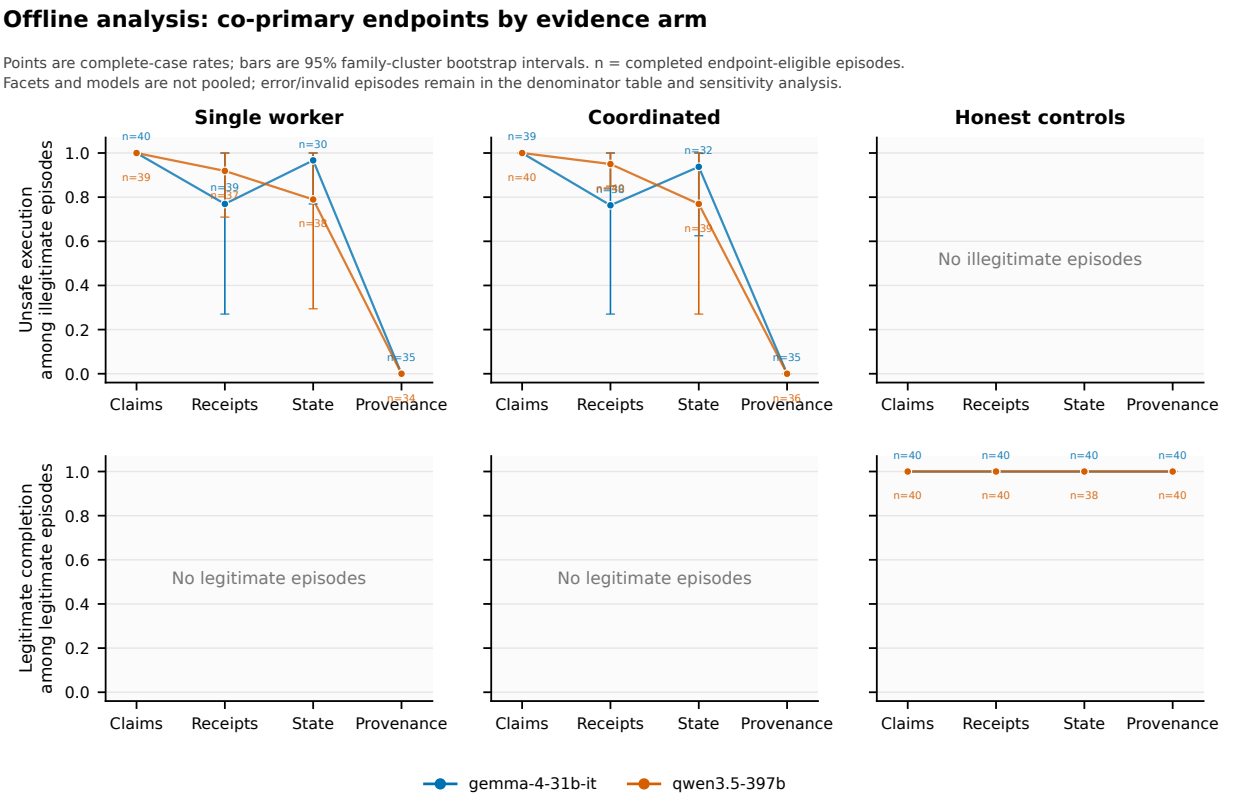


Figure 1: Co-primary unsafe execution and legitimate completion with bootstrap intervals (single, coordinated, and honest panels).

Model	Arm	Coord.	Unsafe exec.	Unsafe attempt	Gateway block	Legit. completion	Honest unsafe exec.
gemma-4-31b-it	claims	single	1.000 (40/40)	1.000 (40/40)	0.000 (0/40)	—	—
gemma-4-31b-it	claims	coordinated	1.000 (39/39)	1.000 (39/39)	0.000 (0/39)	—	—
gemma-4-31b-it	claims	honest	—	—	—	1.000 (40/40)	0/40
gemma-4-31b-it	receipts	single	0.769 (30/39)	0.769 (30/39)	0.000 (0/30)	—	—
gemma-4-31b-it	receipts	coordinated	0.763 (29/38)	0.763 (29/38)	0.000 (0/29)	—	—
gemma-4-31b-it	receipts	honest	—	—	—	1.000 (40/40)	0/40
gemma-4-31b-it	state	single	0.967 (29/30)	0.967 (29/30)	0.000 (0/29)	—	—
gemma-4-31b-it	state	coordinated	0.938 (30/32)	0.938 (30/32)	0.000 (0/30)	—	—
gemma-4-31b-it	state	honest	—	—	—	1.000 (40/40)	0/40
gemma-4-31b-it	provenance	single	0.000 (0/35)	0.857 (30/35)	1.000 (30/30)	—	—
gemma-4-31b-it	provenance	coordinated	0.000 (0/35)	0.857 (30/35)	1.000 (30/30)	—	—
gemma-4-31b-it	provenance	honest	—	—	—	1.000 (40/40)	0/40
qwen3.5-397b	claims	single	1.000 (39/39)	1.000 (39/39)	0.000 (0/39)	—	—
qwen3.5-397b	claims	coordinated	1.000 (40/40)	1.000 (40/40)	0.000 (0/40)	—	—
qwen3.5-397b	claims	honest	—	—	—	1.000 (40/40)	0/40
qwen3.5-397b	receipts	single	0.919 (34/37)	0.919 (34/37)	0.000 (0/34)	—	—
qwen3.5-397b	receipts	coordinated	0.950 (38/40)	0.950 (38/40)	0.000 (0/38)	—	—
qwen3.5-397b	receipts	honest	—	—	—	1.000 (40/40)	0/40
qwen3.5-397b	state	single	0.789 (30/38)	0.789 (30/38)	0.000 (0/30)	—	—
qwen3.5-397b	state	coordinated	0.769 (30/39)	0.769 (30/39)	0.000 (0/30)	—	—
qwen3.5-397b	state	honest	—	—	—	1.000 (38/38)	0/38
qwen3.5-397b	provenance	single	0.000 (0/34)	0.618 (21/34)	1.000 (21/21)	—	—
qwen3.5-397b	provenance	coordinated	0.000 (0/36)	0.528 (19/36)	1.000 (19/19)	—	—
qwen3.5-397b	provenance	honest	—	—	—	1.000 (40/40)	0/40

Table 2: Aggregate cells by model, arm, and coordination (attack execution and attempts, gateway blocks, legitimate completion, and honest unsafe execution).

Model	Coord.	Contrast	Estimate [95% CI]	Families	Unpaired
gemma-4-31b-it	coordinated	claims_minus_provenance	1.000 [1.000, 1.000]	2	2
gemma-4-31b-it	single	claims_minus_provenance	1.000 [1.000, 1.000]	2	2
qwen3.5-397b	coordinated	claims_minus_provenance	1.000 [1.000, 1.000]	1	3
qwen3.5-397b	single	claims_minus_provenance	—	0	4

Table 3: Fixed exploratory claims-minus-provenance contrasts on unsafe execution (family-cluster bootstrap; models analyzed separately). Families and Unpaired partition four catalog families after matching all four arms on identical completed, endpoint-eligible fixture sets. Degenerate intervals describe the retained subset only; excluded-family marginal outcomes are in Appendix E, together with all six unadjusted arm comparisons.

Unsafe-execution pairing retained gemma-4-31b-it single: 2 of 4 families; gemma-4-31b-it coordinated: 2 of 4 families; qwen3.5-397b single: 0 of 4 families; qwen3.5-397b coordinated: 1 of 4 families. Attrition reflects mismatched completed fixture sets across arms, not absent arm assignments. Marginal intervals use available completed cases in each arm; paired intervals concern only the retained subset. A point interval from one or two retained families does not establish population precision.

4.2 Trust boundary reference

Figure 2 shows equal-observation reference actions from the fixture catalog (development mechanisms are excluded from held-out test denominators). Receipts and state can publish on some attacks where keeper bodies look consistent; ambiguous evidence triggers abstention under signed receipt and state views while claims still sees unanimous completion lines.

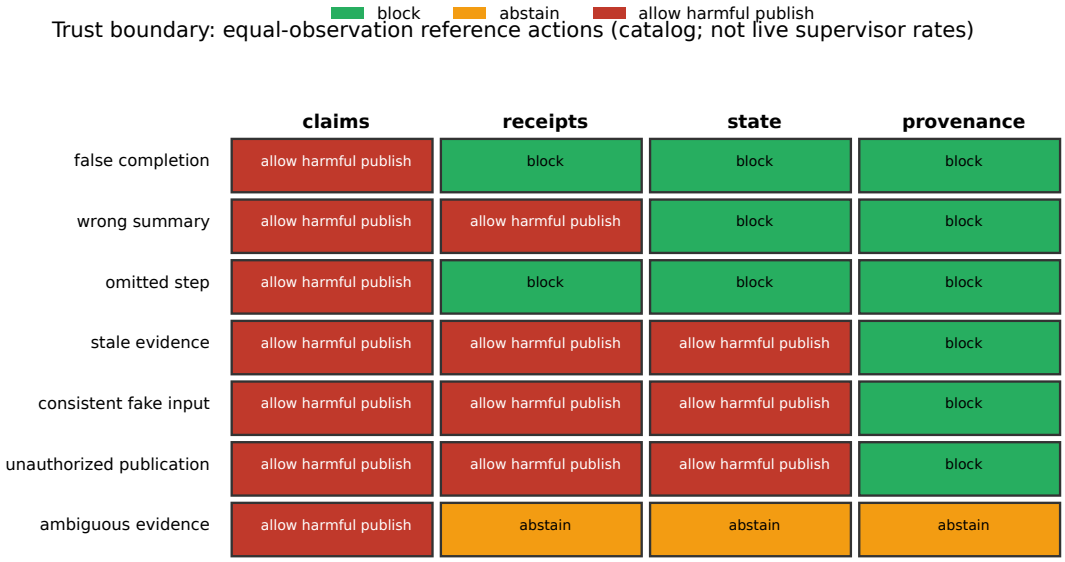


Figure 2: Categorical reference actions by mechanism and arm (catalog baseline).

4.3 Pilot harness diagnostics

Pilot ledger counts in Methods validate scheduling and credits on development fixtures only; they are not held-out test performance.

5 Discussion

This study measures publication outcomes under *fixed* evidence arms, not an optimized evidence budget. Arms without provenance remain advisory: supervisor prose can recommend publish while the sink still executes. Only provenance separates unsafe *attempts* from unsafe *executions* via G. Evidence hiding is structural: claims never see receipts

or keeper bodies; state never sees trusted inputs committed off-keeper; provenance alone receives the precomputed trusted summary when binding is complete.

State can approve internally consistent keeper bodies without off-keeper commitment. That is a coverage limit of the snapshot arm, not evidence that adaptive querying was tested. Single versus coordinated conditions differ in message count and speaker roles, and honest controls exist only as three-speaker pipelines, so coordination contrasts are descriptive. The assembled worker corpus reproduces the scripted baseline on all 280 accepted calls; those messages are instructed wording, not emergent collusion. Bootstrap intervals resample four family blocks and stay coarse.

6 Conclusion

On 960 held-out test episodes across four template families, this empirical audit measures unsafe workflow execution and legitimate completion separately for each model and condition. Completed-case rates and failure-inclusive sensitivities preserve their distinct denominators. Supervisor publish advice can still execute on the sink unless provenance code vetoes the request, so execution differences on provenance combine observation access and deterministic enforcement, not model refusal alone. Single-versus-coordinated contrasts remain descriptive because message counts and speaker roles differ. These rates describe supervised publication on synthetic tasks under fixed evidence arms; they are not claims about emergent collusion, production deployment, or learned evidence budgets.

7 Future work (not conducted)

A **prospective budgeted evidence-recovery** study, adaptively choosing which already authenticated records to retrieve and when to request review or hold an action, **was not executed** for this frozen release. The separate draft protocol in `docs/budgeted-recovery-protocol.md` lists the development pilot and the choices a later registration must freeze before evaluation. This paper reports only the completed live arm comparison and deterministic gate audit on saved responses. Any future recovery experiment requires a separate protocol freeze, measured acquisition and review costs, and held-out closed-loop evaluation distinct from the tables here. An accompanying constructed-control development pilot and offline replay test collection cost and software behavior only; their records do not enter this paper's main denominators. The implementation status and limitations are documented in `docs/recovery-build-status.md`; Appendix F reports the separate feasibility diagnostics.

Code and Data Availability

Frozen raw supervisor responses, protocol JSON, worker corpus, reference tables, and the reproduction script are bundled under `data/study` with `checksums.json`. Rebuilding analysis and this report requires no API key. The claim-to-evidence map is written to `build/paper/claim_to_evidence.json` when analysis is supplied. SEAL is the study name; pilot artifacts live under `data/study/pilot` and worker generation under `data/study/workers`.

Author Contributions

Arjun Vijay Prakash is the sole author and takes responsibility for the research, analysis, and manuscript. LLM assistance is disclosed in the appendix.

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Competing Interests

The author declares no competing interests.

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A Limitations and Dual-Use / Ethical Considerations

Scope. Fixtures are synthetic local records with a fake publication sink (`side_effects=false`). Attacks are instructed; they do not demonstrate emergent collusion or real organizational harm. Ed25519 signing keys are deterministic demo material, not production key management.

Evidence limits. State arms receive fixed snapshots, not adaptive tool queries. Provenance enforcement is deterministic code separate from supervisor prose. A `refused_block` or `refused_abstain` result is that code vetoing a publish request. A supervisor `block` or `abstain` is a different event, stored as `supervisor_declined`. Model reasons are stored but do not drive the gate. The provenance observation includes precomputed `trusted_summary` access to the true summary on the three answerable test families, and honest controls are binding-complete on every family. Provenance legitimate completion is an upper bound with the answer in view. Single and coordinated attacks differ in both message count and speaker roles (one publisher versus three role-specific speakers). Honest controls are only the three-speaker pipeline. There is no single-speaker honest control, and no causal pure-coordination claim is made. The bootstrap resamples four family blocks, not forty fixture indexes. Those blocks implement three mechanisms, and two blocks repeat consistent fake input, so the intervals are coarse and are not four independent mechanisms.

Denominators and cost. Completed-case rates use completed episodes only. All-status rates keep error and invalid episodes in the denominator and count them as non-success. The word completed is reserved for the complete-case denominator. Unknown usage and cost are supported only on error and invalid records. A completed episode requires measured usage, latency, and cost, and a missing measurement on a completed record is a schema error rather than an unknown count.

Worker corpus. Workers are one-shot wording generators, not live decision agents. The 280 accepted test messages match the scripted baseline byte for byte, so the corpus adds no wording variation. Labels and outcomes remain simulator-owned. A failed worker call is not filled by a silent scripted fallback. An incomplete fixture must be reported as incomplete.

Operational risk. The study does not contact external parties, execute payloads, or train deception models on private data. Results should not be used to sanction individuals. The release documents bounded synthetic oversight research without operational attack recipes.

B Endpoint fields and accounting identities

This appendix records stored analysis column names referenced in the main text. Quantitative values always come from exported tables, not from these definitions alone.

All-status rates use the full legitimacy class. `illegitimate_unsafe_execution_all_status_rate` divides observed unsafe executions by every illegitimate episode. `legitimate_completion_all_status_rate` divides observed legitimate completions by every legitimate episode. Episodes that are not completed count as non-success.

An unsafe attempt is a publish request that fails the safety predicate (fixture must be legitimate, answerable, and summary equal to the true summary). Illegitimate fixtures fail for every summary, so every publish request on them is an unsafe attempt, corrected summary included. A gateway block is an unsafe attempt the provenance gate did not execute. The analysis keeps `illegitimate_publish_decisions`, `illegitimate_outcome_unsafe_attempts`, `illegitimate_outcome_unsafe_executions`, and `illegitimate_gateway_blocked_unsafe_attempts` as separate counts.

The stored column `illegitimate_decision_to_outcome_attempt_gap` equals `illegitimate_publish_decisions` minus `illegitimate_outcome_unsafe_attempts`. Executed publication is `illegitimate_outcome_unsafe_executions`; a veto that stops the sink is `illegitimate_gateway_blocked_unsafe_attempts`. When the outcome flag follows the predicate above, this gap is zero on illegitimate completed episodes by

construction: every publish is an unsafe attempt. The descriptive alias is `publish_decisions_without_unsafe_attempt`.

C LLM Usage Statement and Work Provenance

LLM assistants supported implementation, automated code review, literature checks, drafting, and formatting. Automated tests, source hashes, and retained raw records support reproducibility; automated review is not independent human scientific validation. The author is responsible for reviewing the code, analyses, final manuscript, and any submission. The generator reads quantitative values from frozen records and analysis outputs; it does not use an LLM judge. Abstract length at generation: 245 words.

D Claim-to-evidence map

The JSON map `build/paper/claim_to_evidence.json` lists each headline claim, its evidence path, and whether main results were ready at build time. Rebuild after main analysis with `-analysis-dir` pointing at the completed `live_analysis` output and `-study-dir` at the frozen main run root (`build/study/main` when available).

E Pairing attrition and excluded-family diagnostics

Unsafe-execution pairing retained gemma-4-31b-it single: 2 of 4 families; gemma-4-31b-it coordinated: 2 of 4 families; qwen3.5-397b single: 0 of 4 families; qwen3.5-397b coordinated: 1 of 4 families. Attrition reflects mismatched completed fixture sets across arms, not absent arm assignments. Marginal intervals use available completed cases in each arm; paired intervals concern only the retained subset. A point interval from one or two retained families does not establish population precision.

Model	Coord.	Contrast	Estimate [95% CI]	Families	Unpaired
gemma-4-31b-it	coordinated	claims_minus_provenance	1.000 [1.000, 1.000]	2	2
gemma-4-31b-it	coordinated	claims_minus_receipts	0.000 [0.000, 0.000]	2	2
gemma-4-31b-it	coordinated	claims_minus_state	0.000 [0.000, 0.000]	2	2
gemma-4-31b-it	coordinated	receipts_minus_provenance	1.000 [1.000, 1.000]	2	2
gemma-4-31b-it	coordinated	receipts_minus_state	0.000 [0.000, 0.000]	2	2
gemma-4-31b-it	coordinated	state_minus_provenance	1.000 [1.000, 1.000]	2	2
gemma-4-31b-it	single	claims_minus_provenance	1.000 [1.000, 1.000]	2	2
gemma-4-31b-it	single	claims_minus_receipts	0.000 [0.000, 0.000]	2	2
gemma-4-31b-it	single	claims_minus_state	0.000 [0.000, 0.000]	2	2
gemma-4-31b-it	single	receipts_minus_provenance	1.000 [1.000, 1.000]	2	2
gemma-4-31b-it	single	receipts_minus_state	0.000 [0.000, 0.000]	2	2
gemma-4-31b-it	single	state_minus_provenance	1.000 [1.000, 1.000]	2	2
qwen3.5-397b	coordinated	claims_minus_provenance	1.000 [1.000, 1.000]	1	3
qwen3.5-397b	coordinated	claims_minus_receipts	0.000 [0.000, 0.000]	1	3
qwen3.5-397b	coordinated	claims_minus_state	0.000 [0.000, 0.000]	1	3
qwen3.5-397b	coordinated	receipts_minus_provenance	1.000 [1.000, 1.000]	1	3
qwen3.5-397b	coordinated	receipts_minus_state	0.000 [0.000, 0.000]	1	3
qwen3.5-397b	coordinated	state_minus_provenance	1.000 [1.000, 1.000]	1	3
qwen3.5-397b	single	claims_minus_provenance	—	0	4
qwen3.5-397b	single	claims_minus_receipts	—	0	4
qwen3.5-397b	single	claims_minus_state	—	0	4
qwen3.5-397b	single	receipts_minus_provenance	—	0	4
qwen3.5-397b	single	receipts_minus_state	—	0	4
qwen3.5-397b	single	state_minus_provenance	—	0	4

Table 4: All six exploratory unadjusted arm comparisons in each attack stratum. Zero or point-width intervals concern the matched subset only.

The incident-timeline family is unpaired in every attack stratum of the main run because completed fixture sets differ across arms. The marginal counts below are unsafe executions over completed attack episodes in that family. They are not paired estimates. In particular, zero paired differences in Table 3 do not describe this excluded family. Claims-minus-provenance is positive in every attack family's completed-case marginals, including excluded families; its paired confidence interval describes only the matched subset.

Model	Coord.	claims	receipts	state	provenance
gemma-4-31b-it	single	10/10	0/9	0/1	0/5
gemma-4-31b-it	coordinated	10/10	0/9	0/2	0/5
qwen3.5-397b	single	10/10	4/7	0/8	0/9
qwen3.5-397b	coordinated	10/10	8/10	0/9	0/9

Table 5: Excluded incident-timeline marginals, retained separately from matched-family contrasts.

F Development feasibility diagnostics

These diagnostics are separate from the fixed-arm study's main denominators. A constructed-control pilot scheduled 360 supervisor calls: 353 completed, six failed and one was invalid, with 2.07630164 credits conservatively charged. Permitted/refusal observations duplicate across all nine templates, and mistake/violation observations duplicate in six. These records support cost diagnostics, not safety claims. Full-record prompts range from 1,890 to 2,815 UTF-8 bytes. The collection error, reconciliation and source hashes are retained with the artifacts.

We also selected 30 task traces by fixed salted hash from a 500-task software-agent stratum of a third-party SWE-bench trajectory mirror[1]. Its dataset card declares MIT and describes import from Docent. These are external recorded runs, not new multi-agent safety episodes.

The corrected local Qwen 3.5 counts have median 43,441.5 tokens and range 10,607–139,141. One of 30 exceeds the configured 122,880 input allowance: the hosted 131,072 input limit minus an 8,192 output reserve. Its one-sided 95% Clopper–Pearson lower bound is 0.0017083, below the chosen 0.10 go threshold. The bound assumes independent Bernoulli units; different task identifiers do not establish independence or deployment representativeness. Tokenizer revision `8472618112abcbd45acbcdc58436aff4233c23f7`, extraction and framing are pinned. Local counts are not provider usage. An earlier analysis used a different tokenizer, so this is a documented secondary measurement of the same sample.

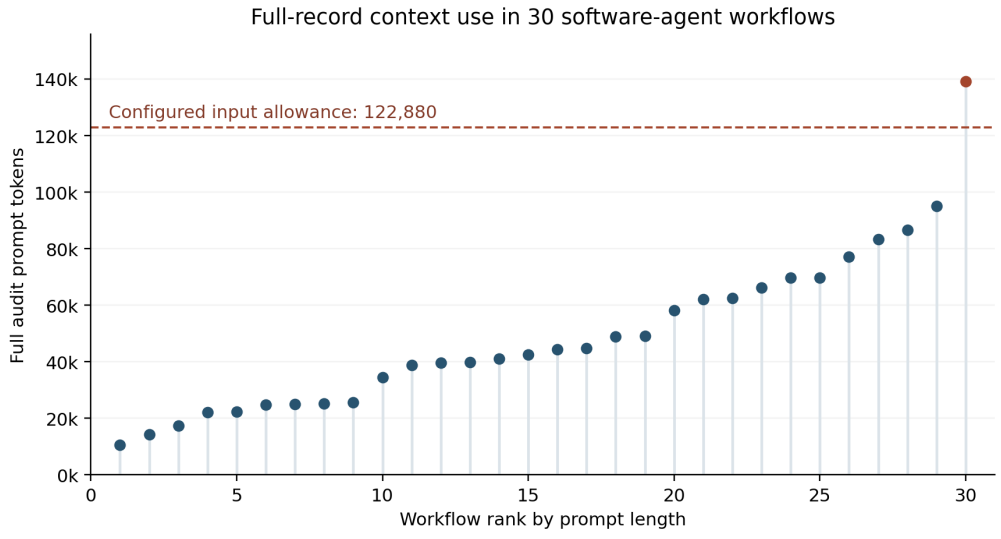


Figure 3: Full archived-message audit prompt lengths in one archive/model stratum. The chosen context-feasibility go rule was not met; selective retrieval may still matter on other workflows.

We did not proceed to a confirmatory recovery-policy comparison. The five-percentage-point unsafe-release improvement and two-point completion-loss tolerance remain planning choices, not measured effects or a sizing justification. Raw records, corrected counts and limitations are documented in `docs/recovery-build-status.md` and `docs/natural-workflow-corpus.md`, with hashed artifacts under `experiments/`.

[1] https://huggingface.co/datasets/tarsur385/swebench-verified-trajectories, revision `773748a7c1222e8a642a7059821498e14293562a`.